

Week 5 Progress Report

Dynamic Row-Level Security in Microsoft Fabric

PROJECT Store Spatial Sales Analytics
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Progress at a glance

This update follows the earlier Week 5 paginated-report submission. It documents the SQL-to-Fabric security-table workflow, dynamic DAX security rule and role assignment, using five screenshots from the actual project. The RLS configuration is set up; end-user enforcement remains unverified because the built-in role test is blocked by Single Sign-On (SSO).

Milestone tracker

Security source table	RegionManagers populated in StoreDB.	EVIDENCED
Fabric pipeline	Copy_RegionManagers added; validated; execution completed (user confirmed).	COMPLETED
Semantic model	RegionManagers visible in existing model.	EVIDENCED
Dynamic RLS rule	Locations filter written using signed-in identity.	CONFIGURED
Role membership	Student account visible in Regional Manager role.	ASSIGNED
Access validation	Test as role unavailable with this SSO setup.	PENDING

Key technical achievement

A reusable email-to-region security mapping has been integrated with the existing StoreDB, Fabric pipeline and Direct Lake semantic model. The design targets region-based visibility without creating separate reports for each manager.

Scope and evidence

Pages 2-4 show the real SQL, Fabric pipeline, semantic model, DAX editor and role-assignment evidence. Screenshots are cropped for readability, and the student email and internal browser URLs are redacted for safer public GitHub posting. The pipeline screenshot was captured while execution was running; its completion was subsequently confirmed by the project owner.

01 | Data preparation and ingestion

From local SQL Server to Microsoft Fabric

The **RegionManagers** table maps an account email to a sales region. The SQL evidence confirms one populated East assignment in StoreDB. Account identifiers have been redacted in the public screenshot.

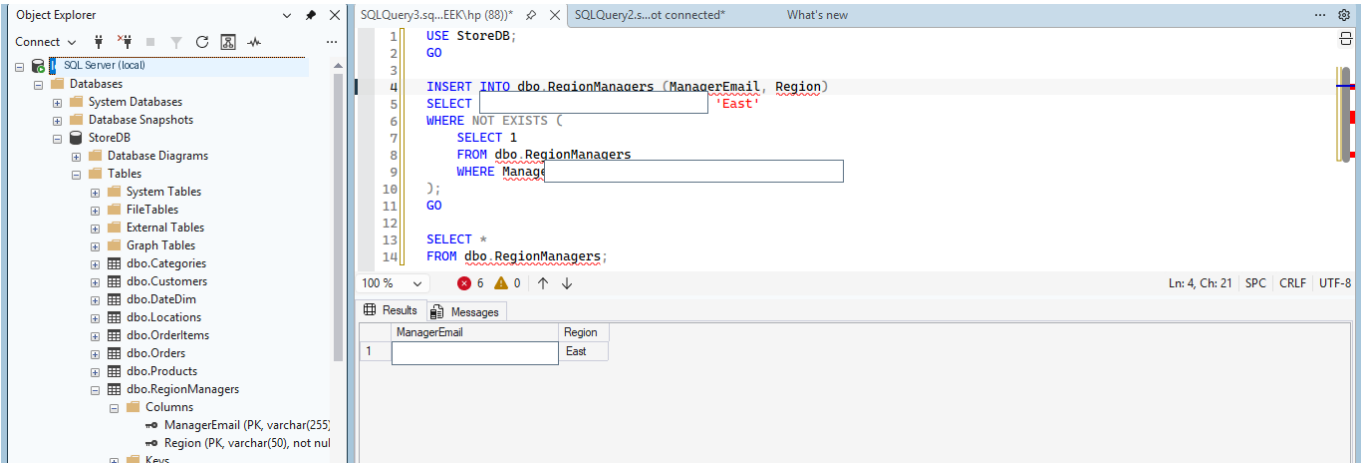


Figure 1. StoreDB security table and its East-region test row, verified in SSMS.

Pipeline integration

The existing **PL_Load_StoreDB** pipeline now includes **Copy_RegionManagers**. The screenshot records successful pipeline validation with no errors. Although the run was marked In progress when the screenshot was captured, the project owner subsequently confirmed that the pipeline finished. The screenshot documents configuration and validation, not the final run-history state.

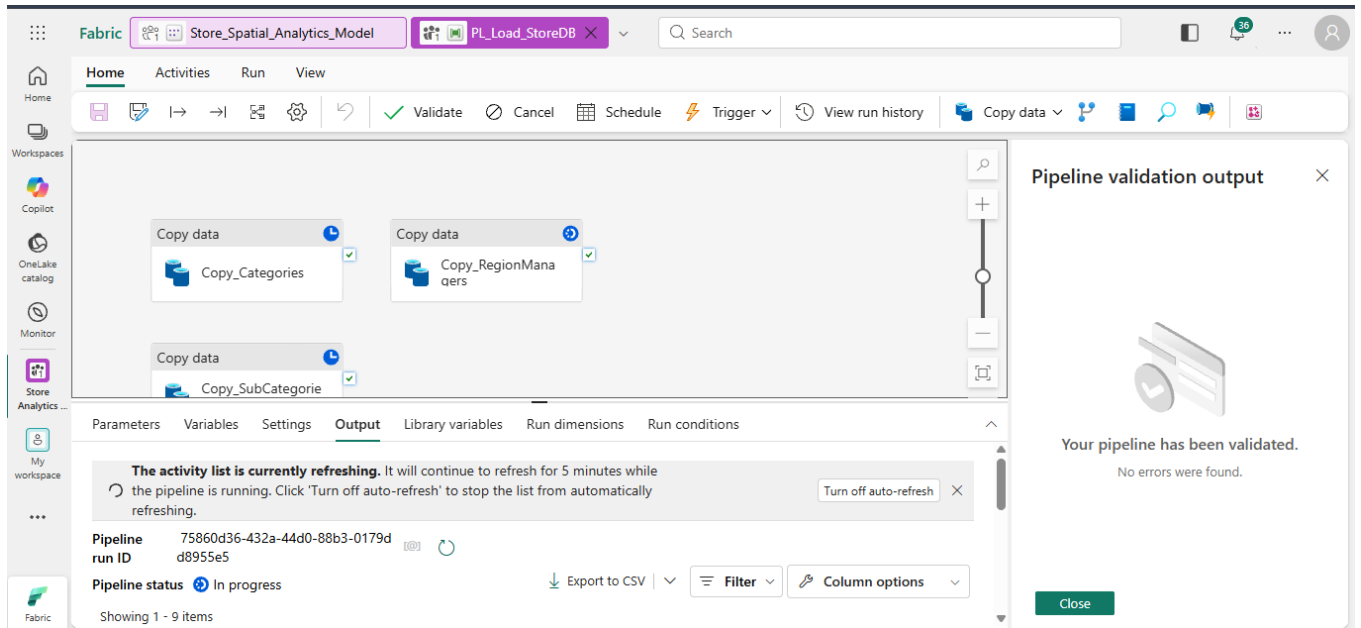


Figure 2. Copy_RegionManagers and successful validation. The screenshot predates the completed pipeline run.

02 | Semantic model and security logic

Dynamic RLS configured in the existing model

The security table was added to **Store_Spatial_Analytics_Model** beside the existing sales, geographic and date tables. In this design, RegionManagers is intentionally disconnected and is referenced by the RLS expression.

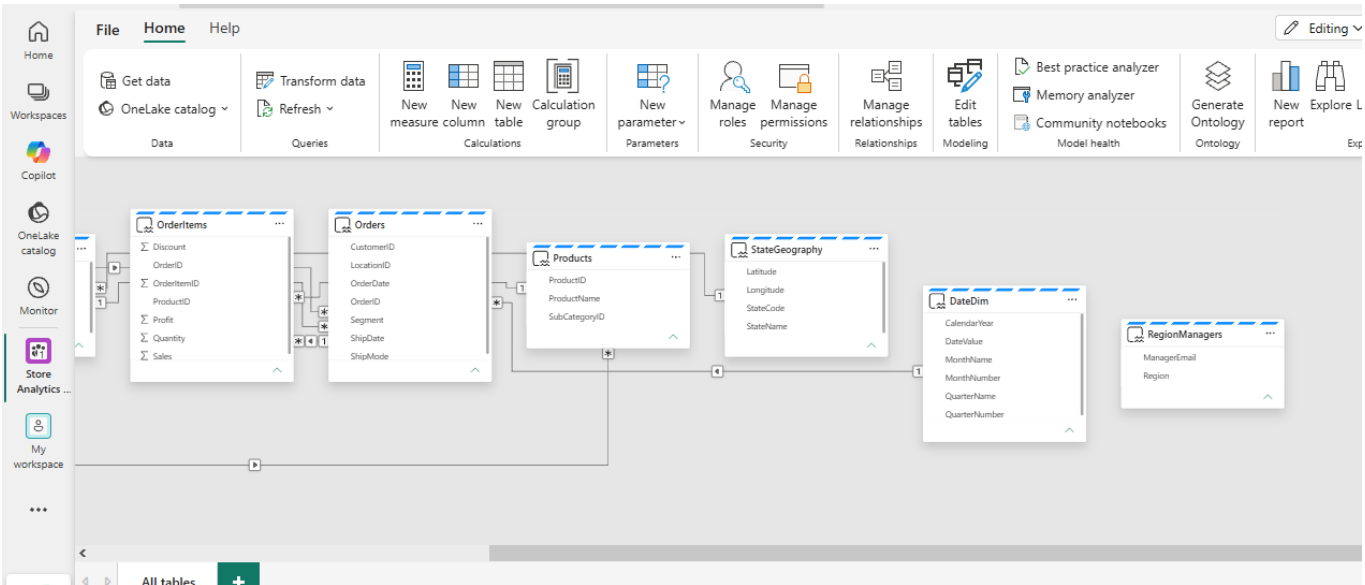


Figure 3. RegionManagers appears as a separate table in the semantic model.

RLS rule in the browser editor

The **Regional Manager** role applies the following Boolean filter to **Locations**. The signed-in user principal name is matched against RegionManagers to identify one assigned territory.

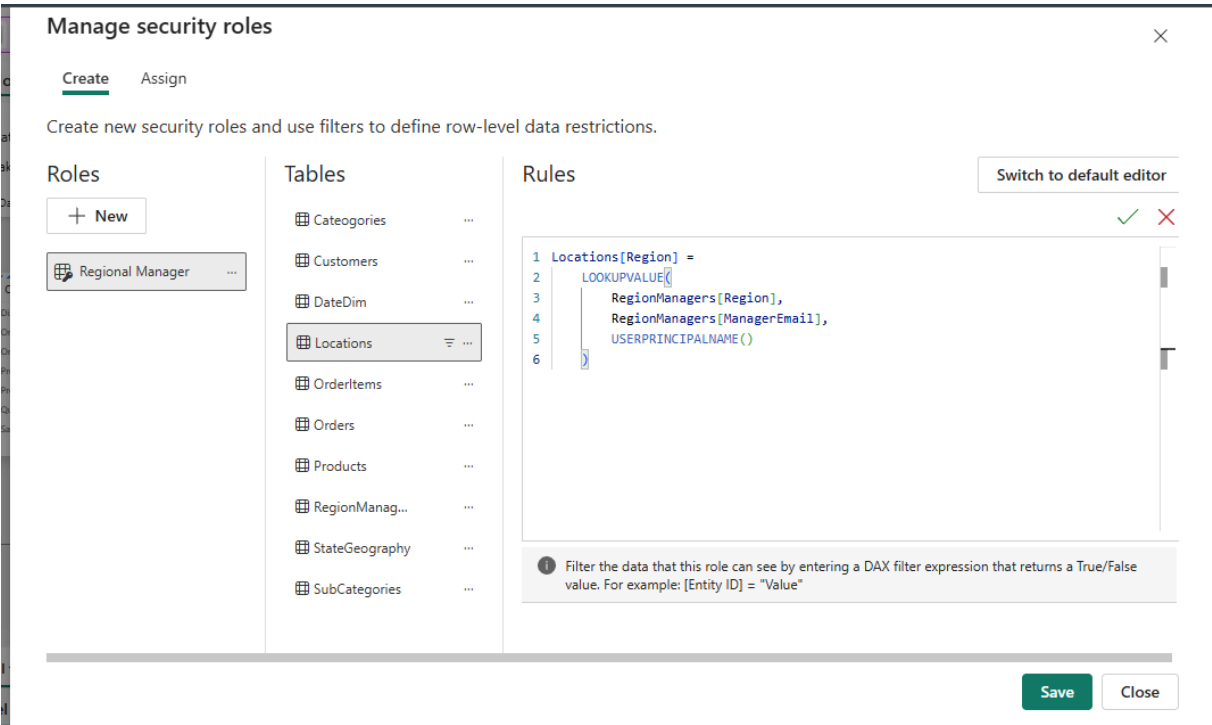


Figure 4. DAX row filter entered on Locations in the Manage security roles editor.

Implementation note: this LOOKUPVALUE design assumes a single region per account. Existing relationships must propagate the Locations restriction through Orders to OrderItems.

03 | Assignment, limitation and next steps

Role membership and test plan

The security screen displays **Regional Manager (1)** with one member assigned. This confirms the role membership is configured; membership alone does not prove that report visibility is correctly restricted.

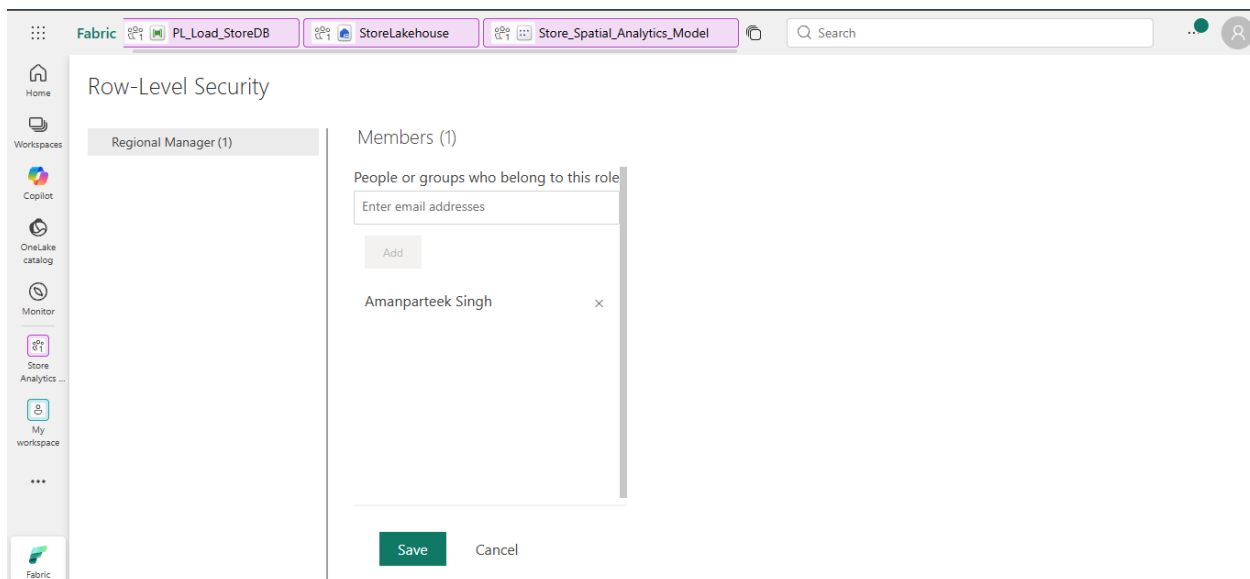


Figure 5. One member assigned to the Regional Manager security role.

Why validation is still pending

Power BI returns “Test as role does not work with Single Sign-On (SSO)” for the current connection. This is a testing limitation, not evidence that the DAX rule is incorrect. A genuine Viewer-account test is needed.

Planned verification and governance

- **Capture completion evidence:** the pipeline has finished; optionally add a final run-history screenshot and check the Lakehouse assignment rows for assessment evidence.
- **Test a real Viewer:** share the report with a separate authorised account assigned in RegionManagers; keep that account out of Admin, Member and Contributor workspace roles.
- **Verify both reports:** clear manual State/Region filters; check that Storereport and the saved Monthly Operational Audit Report show only the authorised territory for the Viewer.
- **Review source access:** avoid unrestricted direct Lakehouse access that could bypass semantic-model-only RLS; document permission decisions before wider sharing.

Evidence boundary

Completed work shown: SQL row, pipeline activity/validation, model table, entered DAX rule and role assignment. Pipeline execution is complete per the project owner, although the supplied screenshot shows an earlier running state. Outstanding: a final run-history screenshot, real Viewer enforcement, paginated-report access test and any automated email distribution. No claim of successful end-to-end RLS verification is made in this report.